

Mohammad Usman Raja

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<https://github.com/usmanrajaa> | <https://leetcode.com/usmanrajaa/>

PORTFOLIO

Personal Portfolio Website: <https://portfolio-usmanrajaas-projects.vercel.app/>

EDUCATION

Institution	Degree	CGPA/Percentage	Duration
IIT Madras	M.Tech, Nanotech	7.63/10	July 2022 – May 2024
St. Aloysius College	B.Sc	62.19%	July 2016 – May 2019
St. Aloysius College	Diploma, Computer Ops	77.90%	July 2017 – May 2018
Higher Secondary Education	12th	86.2%	July 2015 – May 2016
High School Education	10th	90.5%	July 2013 – May 2014

EXPERIENCE

GEEKS OF GURUKUL

Sep. 2024 – Ongoing

Software Developer

- **Automated Answer Evaluation System (Python)-** Developed a Python-based tool to auto-evaluate student answers using text similarity (cosine, fuzzy match), with score generation and analytics reporting.
- **AI:** Developed an OCR model for a student-specific question analysis system and worked on a chatbot that analyzes previous years' questions related to specific subjects.

Code Wisdom

June 2024 – Aug 2024

Data Scientist

- * Developed three AI chatbots designed to assist customers by providing accurate responses based on specific content.
- * These chatbots were built using Retrieval-Augmented Generation (RAG) systems and transformer models for precise and context-aware answers.

Ewebstore, AI SaaS Startup (Nirmaan, IIT Madras)

March 2024 – May 2024

Full Stack Developer Intern

- * **UX/UI + Frontend:** Architected the complete user experience and interface using Figma, and Crafted a fully responsive frontend with React.js.
- * **Backend - AWS:** Prototyped backend logic using Node.js and integrated a PostgreSQL database hosted on AWS S3, enabling seamless API-driven data operations.
- * **AI Component Integration:** Engineered ML models for a no-code platform, automating code generation. Improved development speed by 40% and reduced manual coding errors by 15%.

QuanHack Solutions

May 2024 – June. 2024

Software Developer Intern

- * Developed a cross-platform mobile application using Flutter, adhering to Agile methodologies for rapid and iterative development.
- * Collaborated on RESTful API design and integration to ensure seamless communication between front-end and back-end systems.
- * Configured automated monitoring and performance tracking via Grafana Dashboards, enabling real-time insights into system health.
- * Engineered scalable and secure backend services on Azure, utilizing ADLS Gen2 for storage and SQL for efficient database management.

Electra Wheeler

Jan. 2024 – March 2024

Software Developer Intern

- * Rolled out algorithms such as Dijkstra's for fuel-saving calculators and amortization techniques for EMI calculators, improving calculation accuracy by 20% and supporting precise financial decision-making.
- * Utilized RESTful APIs for subsidy & tax benefit computations and RTO services, ensuring full compliance with regional regulations and reducing user processing time by 30%.

PROJECTS

RGB to Thermal Image Translation

| *VAE, U-Net, Pix2Pix GAN, PyTorch, Image Processing*

Jan. 2024 – Mar. 2024

- * Analyzed 20+ research papers and implemented VAE, U-Net, and Pix2Pix GAN architectures for high-quality image-to-image translation.
- * Achieved a PSNR score of 32.5 using Pix2Pix GAN with a U-Net generator and PatchGAN discriminator.
- * Enhanced performance through data augmentation, advanced loss functions, and optimized weight balancing between adversarial and L1 loss.

TECHNICAL SKILLS

Programming Languages: Python, Java, C++, JavaScript

Frameworks & Technologies: Spring Boot, Hibernate, TensorFlow, PyTorch, Keras, Scikit-learn, OpenCV, NLTK, SpaCy, Cern ROOT, TMVA, Django, React, Node.js

Web Technologies: HTML, CSS, MySQL, SQL, PostgreSQL

Software & Tools: Gradle, Docker, Google Colab, GitHub, AWS, Vertex AI, Grafana, Tableau, Mathematica, MATLAB

COURSES

Advanced & Core Courses: Mathematical Foundation of Data Science, Numerical Techniques Based on C++, Microprocessor and Microcontroller, Quantum Mechanics, Advanced Quantum Mechanics, Deep Learning Masterclass with TensorFlow 2, Supervised Machine Learning, Mathematical Physics, Statistical Physics

POSITIONS OF RESPONSIBILITY

Teaching Assistant: M.Tech (Nanotechnology) Lab at IIT Madras (July – Dec. 2023)

Guided more than 10 students in conducting and understanding advanced nanotech laboratory experiments.

Teaching Assistant: B.Tech Lab at IIT Madras (Jan. – June 2023)

Mentored and supported over 200 students in a key laboratory module, ensuring smooth execution and conceptual clarity.

SCHOLASTIC ACHIEVEMENTS AND PARTICIPATION

GATE (Physics) 2022: Secured an All India Rank of 129 with a 99.15 percentile score.

Competitive Events: Achieved ranks in multiple Vedic Mathematics competitions and district-level model-making competitions.

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